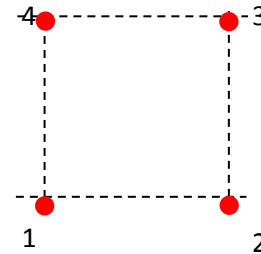


TRABAJO PRÁCTICO Nº 2

CAMPO ELECTROSTÁTICO

P5. Cuatro partículas se emplazan rígidamente en los vértices de un cuadrado de $l = 10\text{cm} = 0,1\text{m}$ de lado. Considere el caso en que todas las partículas están igualmente cargadas con $100\text{ }\mu\text{C}$.



- Realice un gráfico de la situación planteada.
- Determine la dirección y sentido del campo electrostático en el centro del cuadrado y en el punto medio de cada lado sin realizar ningún cálculo.
- Compruebe los resultados obtenidos en el inciso anterior analíticamente.
- Para el caso en que tres partículas tienen cargas positivas y la restante negativa, calcule el campo en el centro del cuadrado y en el punto medio de una arista que tenga como vértice a la carga negativa.

$$\vec{E}_{Pj} = \frac{kQ_j}{(\vec{r}_P - \vec{r}_j)^2} \frac{(\vec{r}_P - \vec{r}_j)}{|\vec{r}_P - \vec{r}_j|}$$

$$\vec{E}_P = \vec{E}_{P1} + \vec{E}_{P2} + \vec{E}_{P3} + \vec{E}_{P4}$$

$$\vec{E}_{P1} = \frac{kQ_1}{(\vec{r}_P - \vec{r}_1)^2} \frac{(\vec{r}_P - \vec{r}_1)}{|\vec{r}_P - \vec{r}_1|}$$

$$\vec{r}_P = (0,0) \quad \vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right)$$

$$\vec{r}_P - \vec{r}_1 = \left(\frac{l}{2}, \frac{l}{2}\right)$$

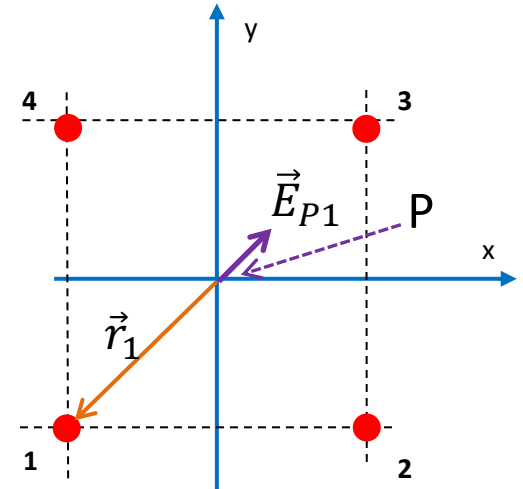
$$|\vec{r}_P - \vec{r}_1| = \frac{l}{\sqrt{2}}$$



$$\frac{\vec{r}_P - \vec{r}_1}{|\vec{r}_P - \vec{r}_1|} = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P1} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{\sqrt{2}}\right)^2} \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P1} = 180 \cdot 10^6 \frac{N}{C} \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$



$$\vec{E}_{P2} = \frac{kQ_2}{(\vec{r}_P - \vec{r}_2)^2} \frac{(\vec{r}_P - \vec{r}_2)}{|\vec{r}_P - \vec{r}_2|}$$

$$\vec{r}_P = (0, 0) \quad \vec{r}_2 = \left(\frac{l}{2}, -\frac{l}{2}\right)$$

$$\vec{r}_P - \vec{r}_2 = \left(-\frac{l}{2}, \frac{l}{2}\right)$$

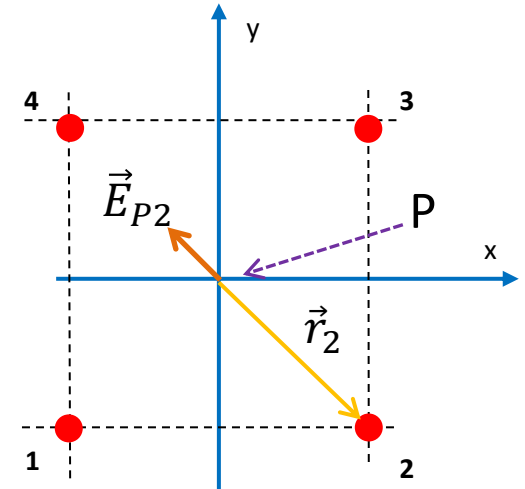
$$|\vec{r}_1 - \vec{r}_2| = \frac{l}{\sqrt{2}}$$



$$\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

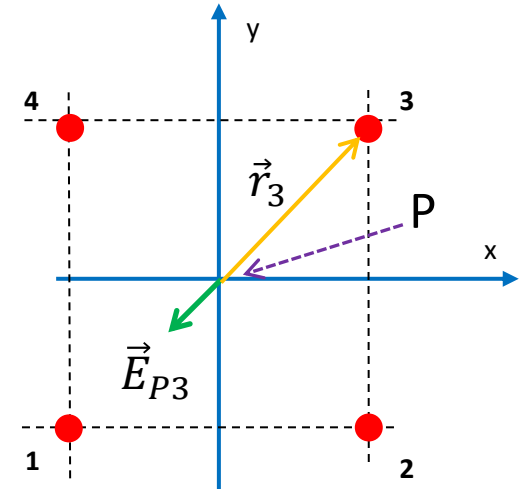
$$\vec{E}_{P2} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{\sqrt{2}}\right)^2} \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P2} = 180 \cdot 10^6 \frac{N}{C} \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$



$$\vec{E}_{P3} = \frac{kQ_3}{(\vec{r}_P - \vec{r}_3)^2} \frac{(\vec{r}_P - \vec{r}_3)}{|\vec{r}_P - \vec{r}_3|}$$

$$\vec{r}_P = (0,0) \quad \vec{r}_3 = \left(\frac{l}{2}, \frac{l}{2}\right)$$



$$\vec{r}_1 - \vec{r}_3 = \left(-\frac{l}{2}, -\frac{l}{2}\right)$$



$$\frac{\vec{r}_1 - \vec{r}_3}{|\vec{r}_1 - \vec{r}_3|} = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$|\vec{r}_1 - \vec{r}_3| = \frac{l}{\sqrt{2}}$$

$$\vec{E}_{P3} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{\sqrt{2}}\right)^2} \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P3} = 180 \cdot 10^6 \frac{N}{C} \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P4} = \frac{kQ_4}{(\vec{r}_P - \vec{r}_4)^2} \frac{(\vec{r}_P - \vec{r}_4)}{|\vec{r}_P - \vec{r}_4|}$$

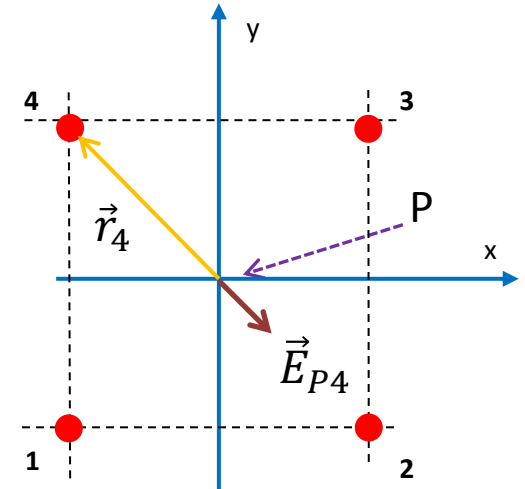
$$\vec{r}_P = (0,0)$$

$$\vec{r}_4 = \left(-\frac{l}{2}, \frac{l}{2}\right)$$

$$\vec{r}_1 - \vec{r}_4 = \left(\frac{l}{2}, -\frac{l}{2}\right)$$

$$|\vec{r}_1 - \vec{r}_4| = \frac{l}{\sqrt{2}}$$

$$\vec{E}_{P4} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{\sqrt{2}}\right)^2} \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$



$$\frac{\vec{r}_1 - \vec{r}_4}{|\vec{r}_1 - \vec{r}_4|} = \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{P4} = 180 \cdot 10^6 \frac{N}{C} \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$\vec{E}_{Pj} = \frac{kQ_j}{(\vec{r}_P - \vec{r}_j)^2} \frac{(\vec{r}_P - \vec{r}_j)}{|\vec{r}_P - \vec{r}_j|}$$

$$\vec{r}_P = \left(0, \frac{l}{2}\right)$$

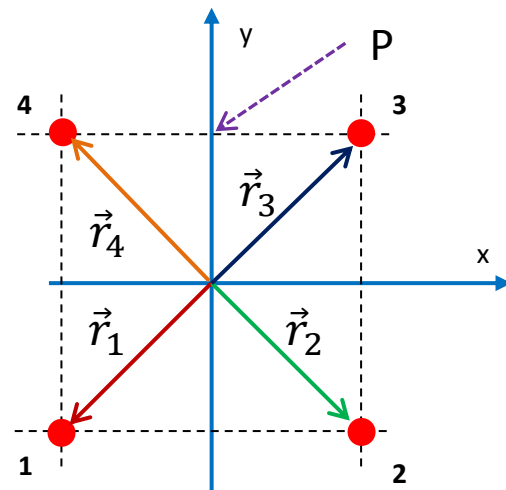
$$\vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right)$$

$$\vec{r}_2 = \left(\frac{l}{2}, -\frac{l}{2}\right)$$

$$\vec{r}_3 = \left(\frac{l}{2}, \frac{l}{2}\right)$$

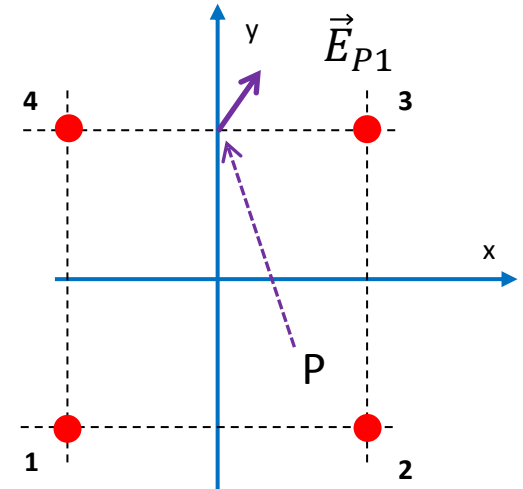
$$\vec{r}_4 = \left(-\frac{l}{2}, \frac{l}{2}\right)$$

$$\vec{E}_P = \vec{E}_{P1} + \vec{E}_{P2} + \vec{E}_{P3} + \vec{E}_{P4}$$



$$\vec{E}_{P1} = \frac{kQ_1}{(\vec{r}_P - \vec{r}_1)^2} \frac{(\vec{r}_P - \vec{r}_1)}{|\vec{r}_P - \vec{r}_1|}$$

$$\vec{r}_P = \left(0, \frac{l}{2}\right) \quad \vec{r}_1 = \left(-\frac{l}{2}, -\frac{l}{2}\right)$$



$$\vec{r}_P - \vec{r}_1 = \left(\frac{l}{2}, l\right)$$



$$|\vec{r}_P - \vec{r}_1| = \frac{5l}{\sqrt{2}}$$

$$\frac{\vec{r}_P - \vec{r}_1}{|\vec{r}_P - \vec{r}_1|} = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) = \left(\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

$$\vec{E}_{P1} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{5 \cdot 0,1m}{\sqrt{2}}\right)^2} \left(\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

$$\vec{E}_{P1} = 7,2 \cdot 10^6 \frac{N}{C} \left(\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

$$\vec{E}_{P2} = \frac{kQ_2}{(\vec{r}_P - \vec{r}_2)^2} \frac{(\vec{r}_P - \vec{r}_2)}{|\vec{r}_P - \vec{r}_2|}$$

$$\vec{r}_P = \left(0, \frac{l}{2}\right) \quad \vec{r}_2 = \left(\frac{l}{2}, -\frac{l}{2}\right)$$

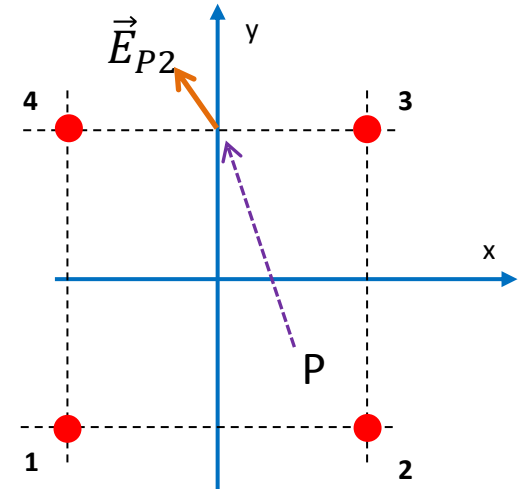
$$\vec{r}_P - \vec{r}_2 = \left(-\frac{l}{2}, l\right)$$

$$|\vec{r}_P - \vec{r}_2| = \frac{5l}{\sqrt{2}}$$



$$\frac{\vec{r}_P - \vec{r}_2}{|\vec{r}_P - \vec{r}_2|} = \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) = \left(-\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

$$\vec{E}_{P2} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{5 \cdot 0,1m}{\sqrt{2}}\right)^2} \left(-\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

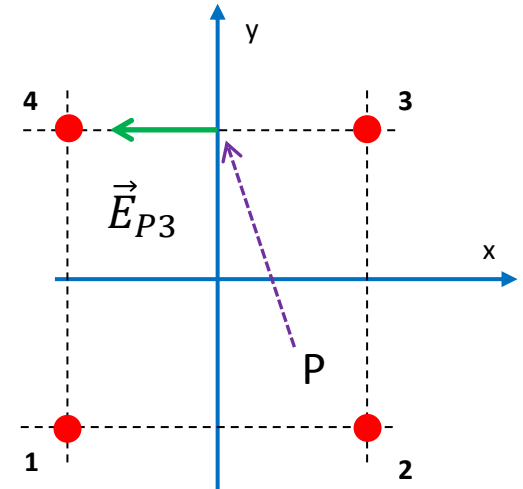


$$\vec{E}_{P2} = 7,2 \cdot 10^6 \frac{N}{C} \left(-\frac{\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$$

$$\vec{E}_{P3} = \frac{kQ_3}{(\vec{r}_P - \vec{r}_3)^2} \frac{(\vec{r}_P - \vec{r}_3)}{|\vec{r}_P - \vec{r}_3|}$$

$$\vec{r}_P = \left(0, \frac{l}{2}\right)$$

$$\vec{r}_3 = \left(\frac{l}{2}, \frac{l}{2}\right)$$



$$\vec{r}_P - \vec{r}_3 = \left(-\frac{l}{2}, 0\right)$$

$$|\vec{r}_P - \vec{r}_3| = \frac{l}{2}$$



$$\frac{\vec{r}_P - \vec{r}_3}{|\vec{r}_P - \vec{r}_3|} = (-1, 0)$$

$$\vec{E}_{P3} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{2}\right)^2} (-1, 0)$$

$$\vec{E}_{P3} = 360 \cdot 10^6 \frac{N}{C} (-1, 0)$$

$$\vec{E}_{P4} = \frac{kQ_4}{(\vec{r}_P - \vec{r}_4)^2} \frac{(\vec{r}_P - \vec{r}_4)}{|\vec{r}_P - \vec{r}_4|}$$

$$\vec{r}_P = \left(0, \frac{l}{2}\right)$$

$$\vec{r}_4 = \left(-\frac{l}{2}, \frac{l}{2}\right)$$

$$\vec{r}_1 - \vec{r}_4 = \left(\frac{l}{2}, 0\right)$$

$$|\vec{r}_1 - \vec{r}_4| = \frac{l}{2}$$



$$\vec{E}_{P4} = \frac{9 \cdot 10^9 \frac{Nm^2}{C^2} 100 \mu C}{\left(\frac{0,1m}{2}\right)^2} (1, 0)$$

$$\vec{E}_{P4} = 360 \cdot 10^6 \frac{N}{C} (1, 0)$$

